17 → 24. find $\boxed{\lim_{x\to 0} \frac{f(a+h)-f(a)}{h}, \; h = \pm 0.1, \pm 0.01}$

17. $f(x) = -7x + 2$, $a = 3$

$$\lim_{x\to 0} \frac{f(3 \pm 0.1) - f(3)}{0.1}$$

⊕

$$\lim_{x\to 0} \frac{f(3+0.1) - f(3)}{0.1}$$

$$\frac{f(3.01) - f(3)}{0.1}$$

$$= \frac{-19.7 + 19}{0.1}$$

$$= -7$$

$$\lim_{h\to 0^+} f(x) = -7$$

⊖

$$\lim_{x\to 0} \frac{f(3-0.1) - f(3)}{-0.1}$$

$$\frac{-18.3 + 19}{-0.1} = -7$$

$$\lim_{h\to 0^-} f(x) = -7$$

$$\Longrightarrow \boxed{\lim_{h\to 0} f(x) = -7}$$

18. $f(x) = 9x + 0.06$, $a = -1$

$$\lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \rightarrow f(-1) = -8.94$$

⊕

$$\frac{f(-1+0.1) \overset{+8.94}{-\cancel{f(-1)}}}{0.1}$$

$$\frac{-8.04 + 8.94}{0.1} = 9$$

$$\lim_{h\to 0^+} f(x) = 9$$

⊖

$$\frac{f(-1-0.1) \overset{+8.94}{-\cancel{f(-1)}}}{0.1}$$

$$\frac{-9.84 + 8.94}{-0.1} = 9$$

$$\lim_{h\to 0^-} f(x) = 9$$

$$\lim_{h\to 0} f(x) = 9$$

19. $f(x) = x^2 + 3x - 7$, $a = 1$

$$\lim_{h\to 0} \frac{f(1 \pm 0.1) - f(1) \rightarrow (-3)}{\pm 0.1}$$

⊕

$$\frac{f(1+0.1) - f(1)}{0.1}$$

$$= \frac{\overset{-2.49}{\cancel{((1.1)^2 + 3(1.1) - 7)}} - \overset{-3}{\cancel{((1)^2 + 3(1) - 7)}}}{0.1}$$

$$= \frac{-2.49 + 3}{0.1} = 5.1$$

$$\lim_{h\to 0^+} f(x) = 5.1$$

⊖

$$\lim_{h\to 0} \frac{f(1-0.1) - f(1)}{-0.1}$$

$$\frac{-3.49 + 3}{-0.1} = -4.9$$

$$5.1 \neq 4.9$$

$$\Rightarrow \lim_{h\to 0} f(x) = \text{DNE} \quad X$$

20. $f(x) = \frac{1}{x+1}$, $a = 2 \rightarrow 0.33$

$$\lim_{h\to 0} \frac{f(2 \pm 0.1) - f(a)}{0.1}$$

⊕

$$\lim_{h\to 0} \frac{f(2+0.1) - 0.\overline{33}}{0.1}$$

$$\approx \frac{0.323 - 0.\overline{333}}{0.1} = \frac{-0.010752}{0.1}$$

$$= \boxed{-0.107}$$

$$\lim_{h\to 0^+} f(x) = -0.107$$

⊖

$$\lim_{h\to 0} \frac{f(2-0.1) - f(2) \rightarrow 0.\overline{333}}{-0.1}$$

$$= \frac{3.448 - 0.\overline{33}}{-0.1}$$

$$= \frac{0.01149}{-0.1} \approx 0.1149$$

$$\lim_{h\to 0} f(x) \approx -0.1149$$

$$-0.107 \neq -0.1149$$

$$\lim_{h\to 0} f(x) = \text{DNE} \quad X$$

22. $f(x) = \ln x$, $a = 5$

$$\lim_{h\to 0} \frac{f(5 \pm 0.1) - f(5)}{\pm 0.1}$$

⊕

$$\lim_{h\to 0} \frac{f(5+0.1) - f(5)}{0.1}$$

$$f(5) = 1.609$$

$$\frac{f(5.1) - f(5)}{0.1} = \frac{1.629 - 1.609}{0.1}$$

$$= \boxed{0.198}$$

$$\lim_{h\to 0^+} f(x) = 0.198$$

⊖

$$\lim_{h\to 0^-} \frac{f(5-0.1) - f(5)}{-0.1}$$

$$\frac{1.589 - 1.609}{-0.1} = -0.2$$

$$\lim_{h\to 0^-} f(x) = -0.2$$

$$0.198 \neq -0.02$$

$$\lim_{h\to 0} f(x) = \text{DNE} \quad X$$

23. $f(x) = \sin x$, $a = \pi$

$$\lim_{h\to 0} \frac{f(\pi \pm 0.1) - f(\pi)}{\pm 0.1}$$

⊕

$$\lim_{h\to 0^+} \frac{f(\pi+0.1) - f(\pi)}{+0.1}$$

$$f(\pi + 0.1) = 0.056$$

$$f(\pi) = 0$$

$$= \frac{0.056 - 0}{0.1} = 0.56$$

$$\lim_{h\to 0^+} f(x) = 0.56$$

⊖

$$\lim_{h\to 0^-} \frac{f(\pi - 0.1) - f(\pi)}{-0.1}$$

$$f(\pi - 0.1) = 0.053$$

$$f(\pi) = 0$$

$$= \frac{0.053 - 0}{-0.1} = -0.53$$

$$\lim_{h\to 0^-} f(x) = -0.53$$

$$0.56 \neq 0.53$$

$$\lim_{h\to 0^-} f(x) = \text{DNE} \quad X$$